

CSCE 679 DATA VISUALIZATION · MODULE 01 · LECTURE NOTES

Visualisation as Encoding

The mapping decides what can be read.

MODULE OUTCOMES

- Define a visualisation as a data-to-channel mapping.
- State the what-why-how framing and use it.
- Explain why the encoding determines the readable questions.
- Explain what visualisation is and is not good for.
- Specify an encoding rather than choosing a chart type.

1 The mapping

A VISUALISATION IS A SET OF MARKS WHOSE VISUAL CHANNELS ENCODE DATA ATTRIBUTES

A **mark** is a geometric primitive — a point, a line, an area, a volume — and a **channel** is a visual property of that mark which can be varied: position, length, angle, slope, area, colour hue, colour luminance, shape, texture, or motion.

So a scatter plot is: point marks, with horizontal position encoding one variable and vertical position encoding another — and a bar chart is line or area marks with length encoding the value and position encoding the category. Both are mappings, described completely by their mark and their channel assignments.

Which means 'bar chart' and 'scatter plot' are names for common mark-and-channel combinations rather than primitives — and thinking in combinations rather than in named charts is what lets you design a figure nobody has named, which is frequently what unusual data requires.

And the channel choice is the design decision, because channels differ enormously in how accurately they are read (Module 02) — which is measured rather than asserted, and is what makes this subject empirical rather than a matter of style.

INSTEAD OF "a grouped bar chart", write:

```
mark:    bar
x:       month      (ordered key)
y:       revenue    (quantitative value)
colour:  product line (categorical key)
```

WHICH MAKES THE DECISIONS VISIBLE

```
why is product line on colour rather than on a
  second axis?
why is revenue on length rather than on area?
is month ordered, and does the encoding respect
  that order?
```

AND IT MAKES THE ALTERNATIVES ENUMERABLE

```
put product line on a small-multiple facet
put revenue on position rather than on length
put month on a continuous axis as a line
```

SO: specify the encoding, then look at the chart it produces -- rather than choosing the chart and inheriting its encoding.

Specify the encoding and inherit the chart — rather than choosing the chart and inheriting its encoding, which is the usual direction and which hides every decision inside a chart-type name. This inversion is the module's practical contribution, and it is why the course recommends a grammar-of-graphics library: the grammar forces the specification, where a chart-type API supplies it silently.

2 What, why, how

Question	What it asks	Why it comes first
What	What is the data? Attribute types, cardinality, and structure.	It constrains which channels are even available (Module 03) — an unordered category cannot go on a quantitative channel.
Why	What task is the reader performing?	It determines which channel has to be accurate — see the callout.
How	The marks, the channels, and the interaction.	It follows from the other two, and from nothing else.

'How' is the only one most people start with — the question 'what chart should I make' is a 'how' question asked first — and a chart chosen before the reader's task is known is optimised for nothing in particular, which is exactly why it looks fine and reads badly. The ordering is the whole discipline of this framework, and it costs nothing but the habit.

THE READER'S TASK DETERMINES WHICH CHANNEL HAS TO BE ACCURATE

Looking up one value, comparing two values, finding the extreme, judging a trend, estimating a total, and judging a proportion are all different tasks — and an encoding that is good for one is frequently poor for another.

A pie chart supports 'is this roughly half' and fails badly at 'which of these two similar slices is larger'; and a stacked area chart supports judging the total while making the individual components hard to compare, where a grouped bar chart does the reverse. Neither is wrong; they answer different questions.

So naming the task is what makes an encoding arguable — 'this chart is bad' is a preference, and 'this chart does not support the comparison the reader needs' is a claim that can be tested (Module 13 §1).

And a figure supporting several tasks at once is a compromise, which is frequently correct and should be a decision rather than an accident — or should be resolved by showing several linked views (Module 06) rather than by overloading one.

3 What visualisation is for

- **It reveals structure nobody specified** — clusters, outliers, gaps, discontinuities, and shapes that no summary statistic names — **which is the strongest argument for it** and is the callout below.
- **It supports comparison across many items simultaneously**, where a table requires serial reading and therefore holding values in working memory (CSCE 671 Module 03 §2's four-chunk limit).
- **And it uses the visual system's parallelism**, which locates an item differing in one channel in approximately constant time regardless of how many others there are (CSCE 671 Module 02 §3's preattentive features).
- **But a table is better for looking up exact values** — reading a number off an axis is slow and approximate — and **a single number is better when there is genuinely one number**, which is more often than the number of charts produced would suggest.
- **So: a chart for pattern, a table for lookup, and a sentence for a conclusion** — and **most reports need all three** rather than one of them repeated. **A figure used for lookup is a table drawn badly**, and is a common and avoidable mistake.

AND THE STRONGEST CASE FOR VISUALISATION IS THE STRUCTURE A STATISTIC HIDES

Datasets with identical means, variances, and correlation coefficients can have completely different shapes — one linear, one strongly curved, one a tight cluster with a single outlier dragging the fitted line — and **the summary statistics cannot distinguish between them at all**.

Which is a constructed demonstration (Anscombe's quartet, and its modern elaborations) and also a real phenomenon: the same thing happens in real data whenever a relationship is non-linear or a subgroup behaves differently from the aggregate (CSCE 638 Module 12 §1's disaggregation argument, arriving visually).

So visualisation is part of the analysis rather than a presentation of it — which is **CSCE 676 Module 01's two-problem framing with the picture serving as a tool for the second problem**: deciding whether what you found is real.

And it is why 'look at the data first' is methodological advice rather than aesthetic — **a model fitted without looking is fitted to something you did not inspect**, and the inspection is cheap.

4 Reading this course

- **Modules 02 to 05 are the measurements** — the channel accuracy ranking, the data types and what channels suit them, colour, and the chart types that follow from all three.
- **Modules 06 to 08 are the hard structures** — multiple coordinated views and interaction, networks and hierarchies, and geographic data — all of which have structure the plane does not naturally hold.
- **Module 09 is uncertainty**, which is **the most often omitted and most often decisive** element of a figure, and which has its own module for that reason.

- **Modules 10 to 12 are deception, scale, and scientific visualisation** — the last of which **connects directly back to CSCE 647's volume rendering and ray marching**, which is a pleasant closing of a loop in this track.
- **And Module 02 is the one to internalise**, because **the channel ranking turns chart choice from taste into prediction**. If you take one thing from this course, take the channel ranking — it is short, it is empirical, and it **decides most design questions before they become arguments**.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Munzner — Visualization Analysis and Design, chapters 1 through 3 (slides free)	§1 and §2 — the what-why-how framework and the mark-and-channel vocabulary, from the source.
Wilke — Fundamentals of Data Visualization, chapter 1 (free)	§1's mapping framing , with the aesthetics-as-channels vocabulary used throughout the book.
Anscombe — Graphs in Statistical Analysis (free)	§3's callout in the original — two pages, and it makes the argument once and permanently.
Matejka & Fitzmaurice — Same Stats, Different Graphs (free)	§3 extended — arbitrary shapes with identical statistics, generated, which is more striking than the original.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write the encoding specification** for five charts you did not make.
- 2 **Enumerate three alternative encodings** for one of them.
- 3 **Take a chart you made** and write down the reader's task it was for.
- 4 **Find a chart** where the task and the encoding do not match.
- 5 **Find a figure used for lookup** and replace it with a table.
- 6 **Find a chart that should have been one number**.
- 7 **Construct two datasets** with the same mean and correlation and different shapes.
- 8 **Plot both** and confirm the statistics do not distinguish them.
- 9 **Fit a linear model to each** and report what you would have concluded without looking.
- 10 **Write the three-question list** (lookup, comparison, trend) for Project 1's dataset.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define a mark and a channel, and give the encoding of a scatter plot.
- 2 Why is 'bar chart' not a primitive?
- 3 What does specifying the encoding make visible?
- 4 Give the three framework questions and why the order matters.
- 5 Why does the reader's task determine the encoding?
- 6 Contrast what a pie chart and a bar chart support.
- 7 What does visualisation do that a table does not, and vice versa?
- 8 State the one-sentence division of labour.
- 9 Why can identical statistics have different shapes, and what follows?
- 10 Which module is the one to internalise, and why?